Hall Effect in Semiconductors

Figures Compilation

All figures generated by hall_effect_analysis.py

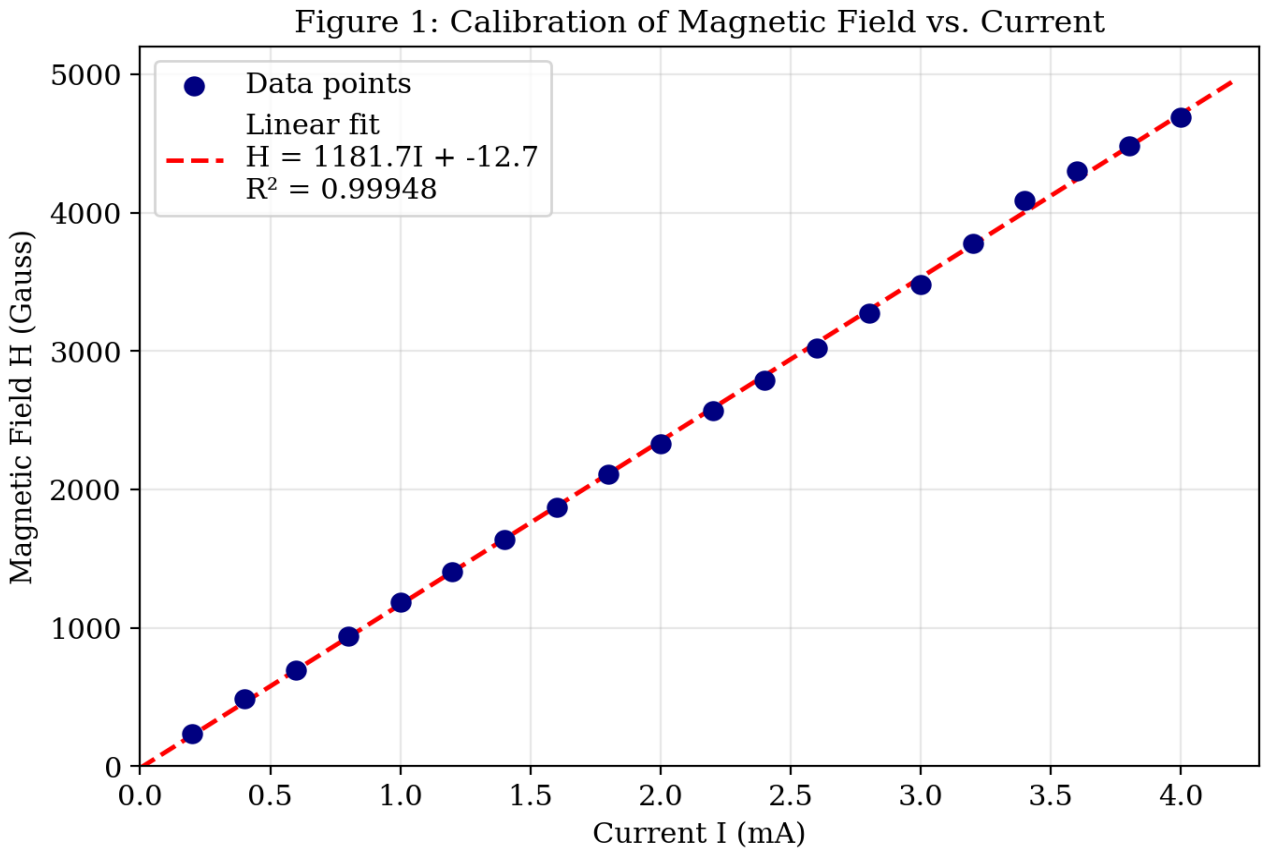

Figure 1: Magnetic Field Calibration — Corrected magnetic field H_{corr} as a function of electromagnet current I_M. Linear fit: H = 1181.67·I – 12.66 Gauss (R^2 = 0.9995).

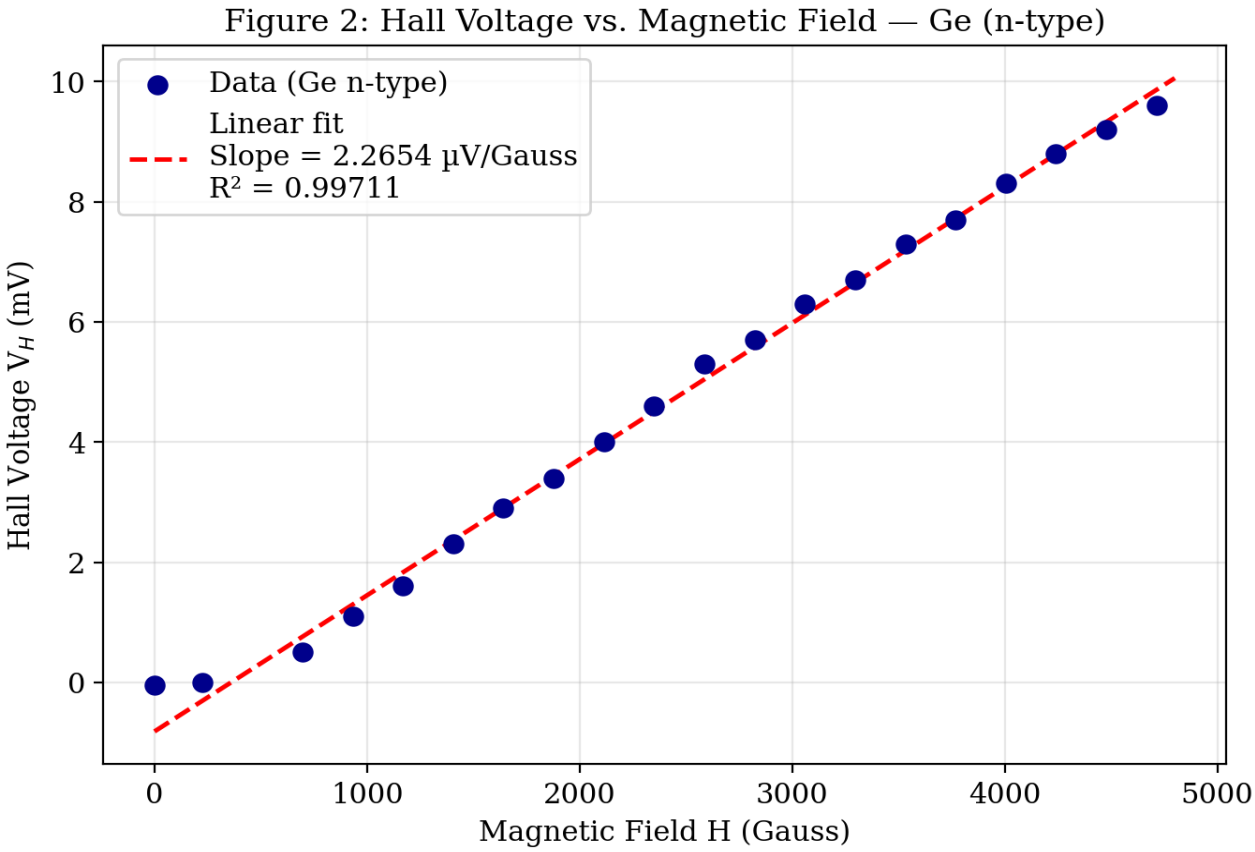

Figure 2: Hall Voltage vs. Magnetic Field — Ge (n-type), probe current I_p = 0.49 mA. Slope dV_H/dH = 2.27 µV/Gauss, R^2 = 0.997.

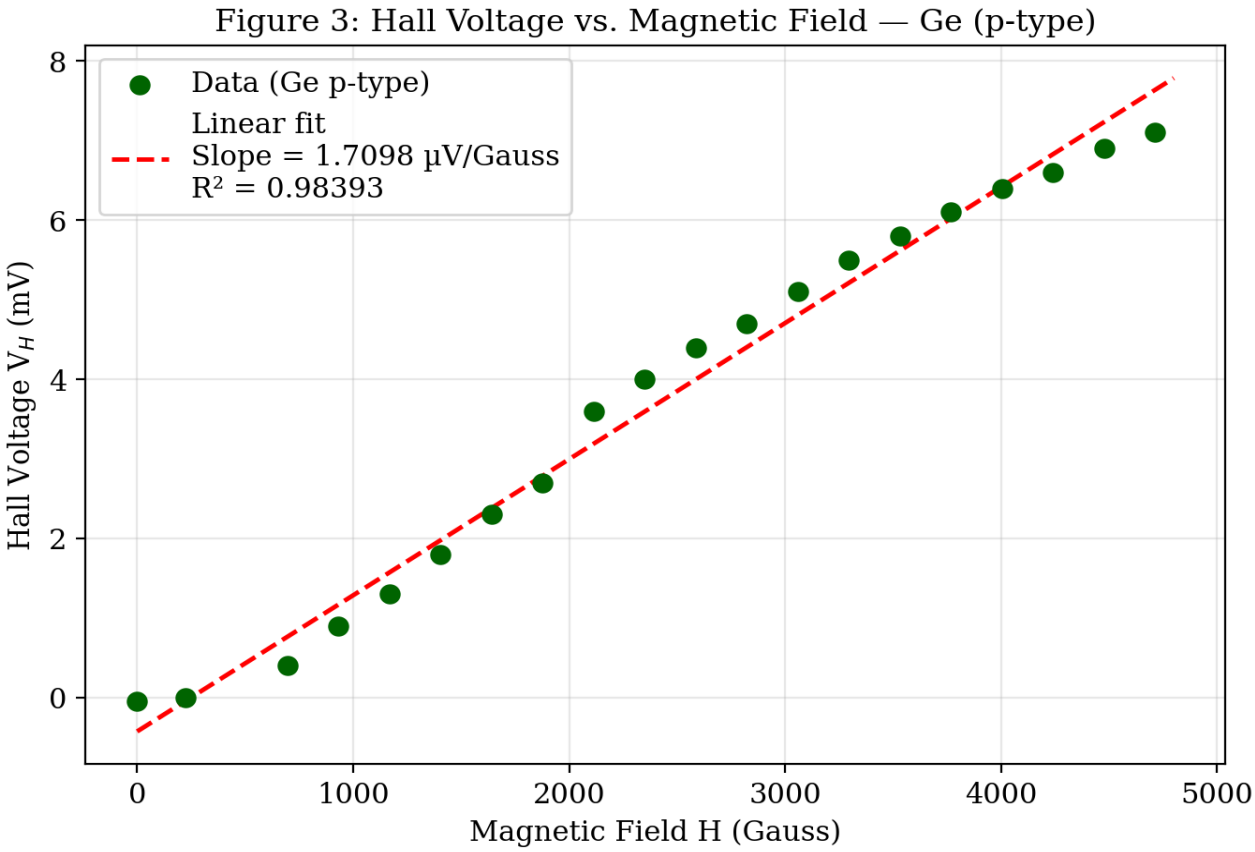

Figure 3: Hall Voltage vs. Magnetic Field — Ge (p-type), probe current I_p = 0.50 mA. Slope dV_H/dH = 1.71 µV/Gauss, R^2 = 0.984.

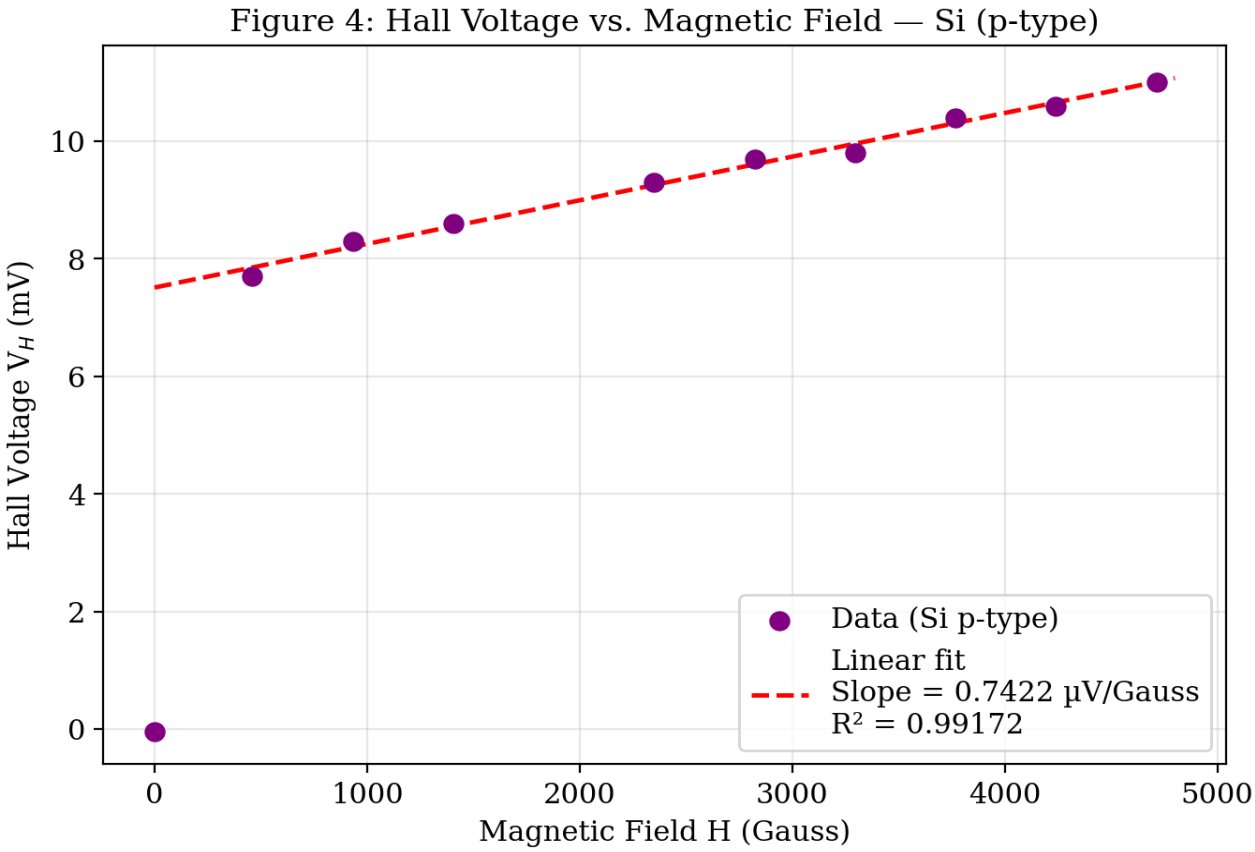

Figure 4: Hall Voltage vs. Magnetic Field — Si (p-type), probe current I_p = 0.45 mA. Slope dV_H/dH = 0.74 μV/Gauss, R^2 = 0.992.

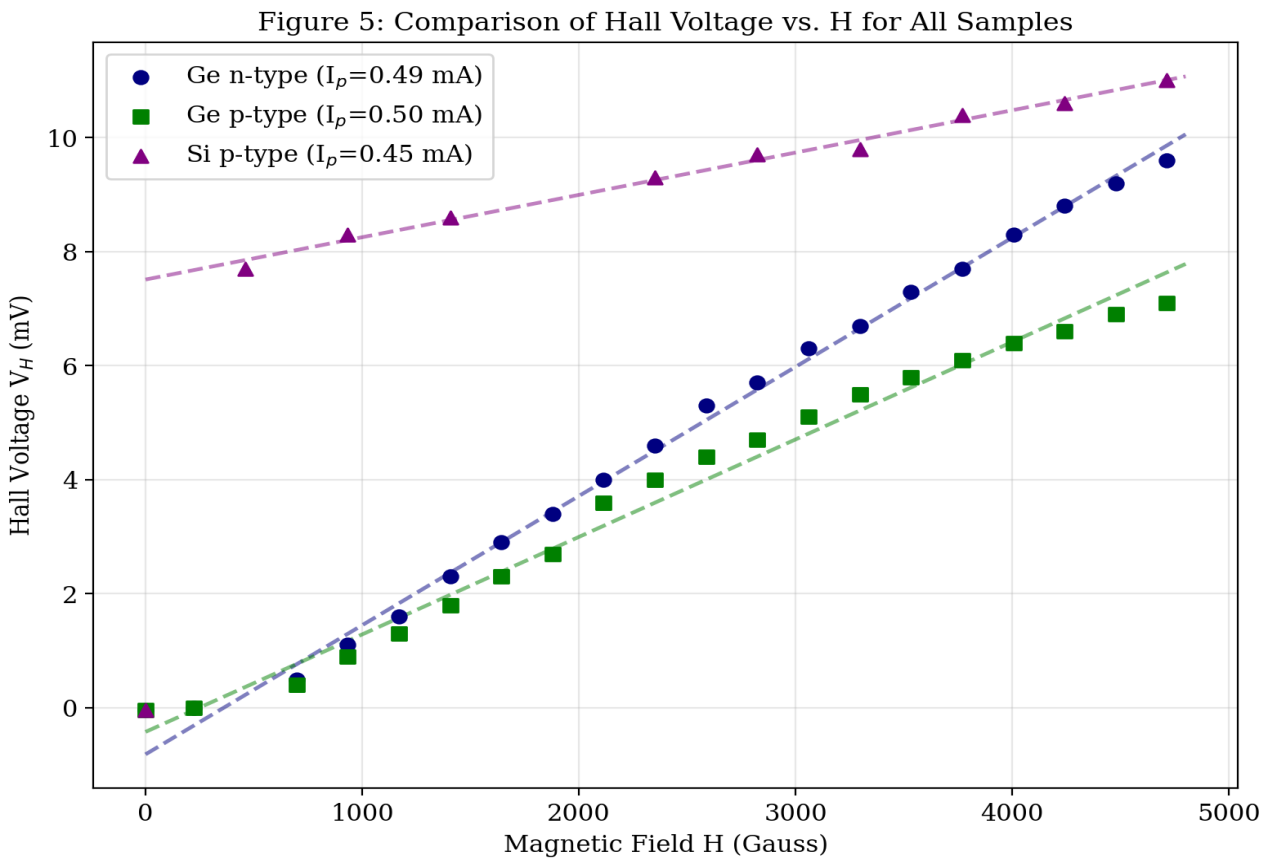

Figure 5: Comparison of Hall Voltage vs. Magnetic Field for all three samples. Ge (n-type) shows the steepest slope owing to higher electron mobility.

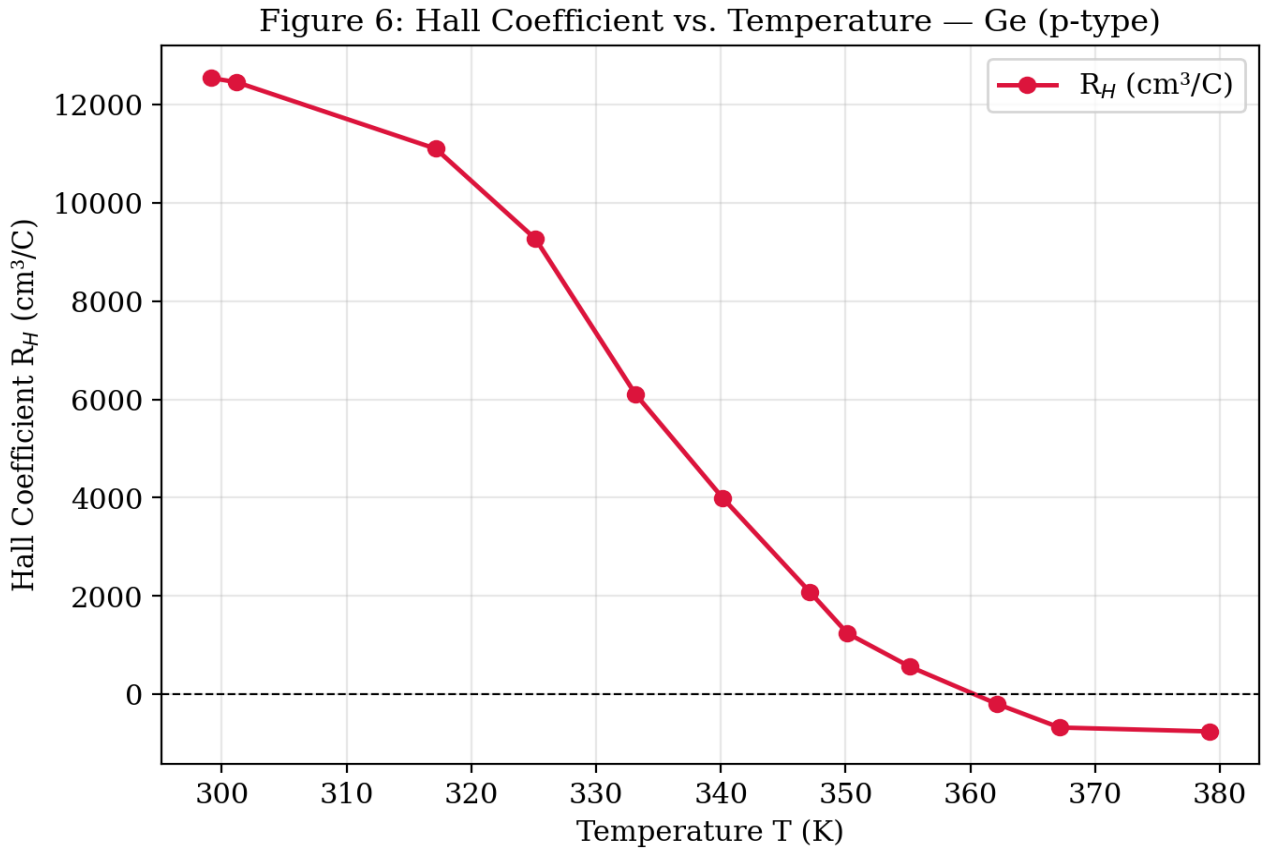

Figure 6: Hall Coefficient R_H vs. Temperature — Ge (p-type). The sign change near 360 K marks the onset of intrinsic conduction.

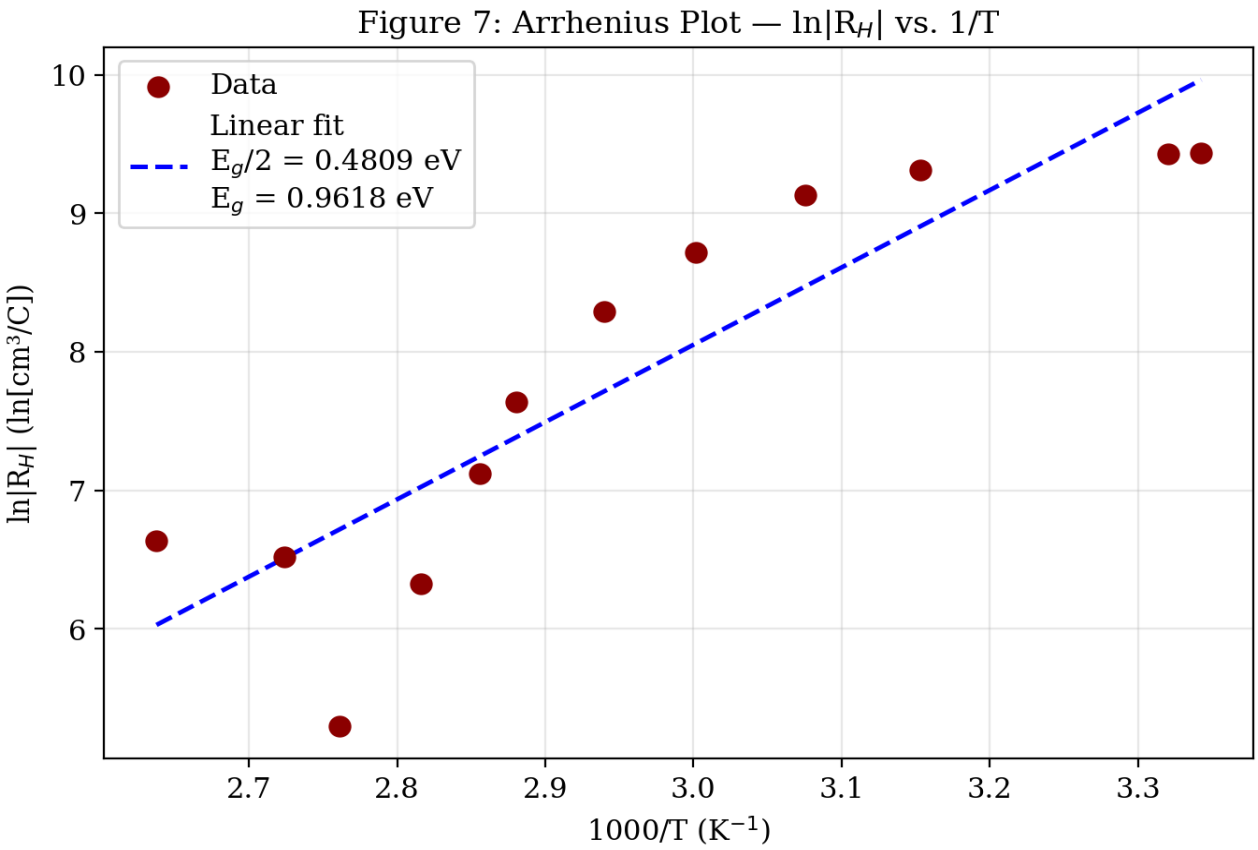

Figure 7: Arrhenius Plot — $\ln|R_H|$ vs. 1/T for Ge (p-type). Linear fit gives slope ≈ 11150 K → E_g ≈ 0.96 eV.

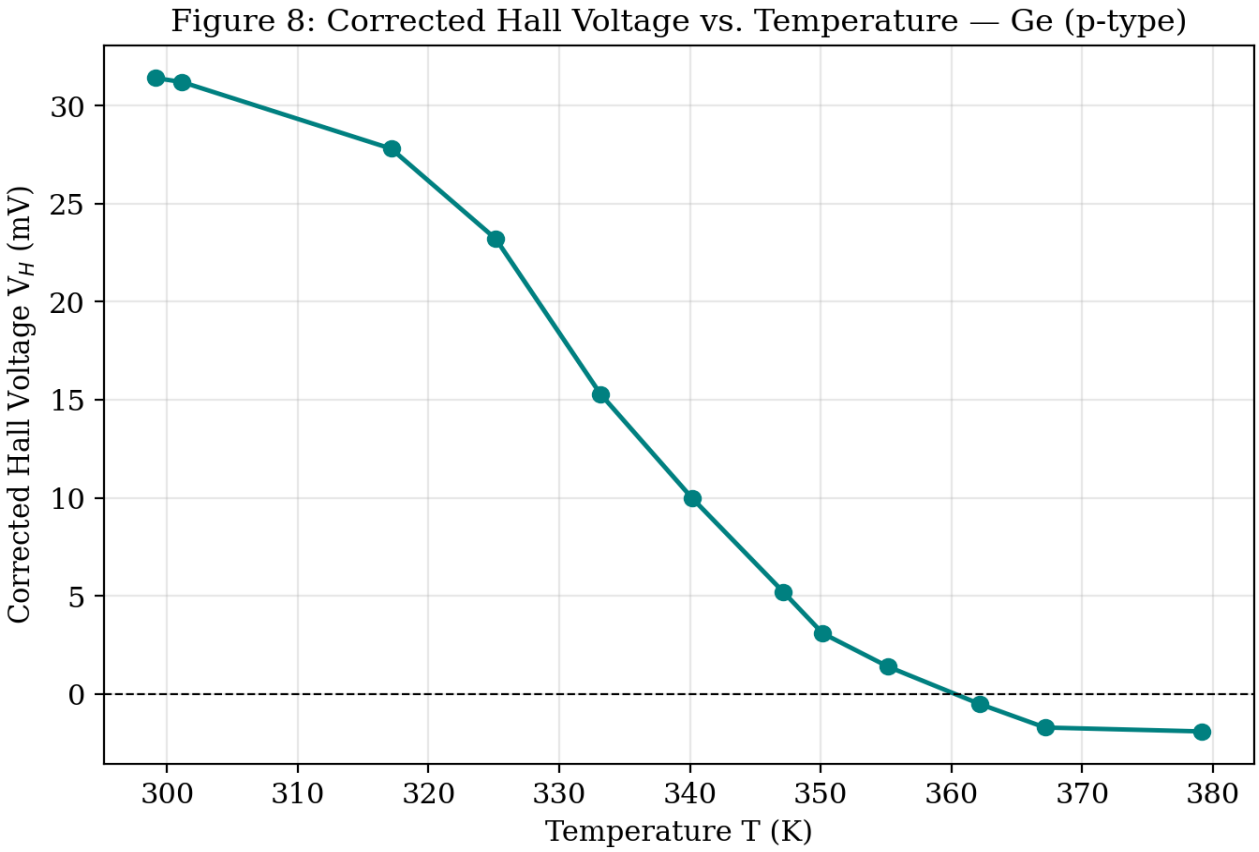

Figure 8: Corrected Hall Voltage V_H vs. Temperature — Ge (p-type), B = 3130 Gauss, I_p = 4.00 mA. Zero crossing corresponds to carrier compensation.

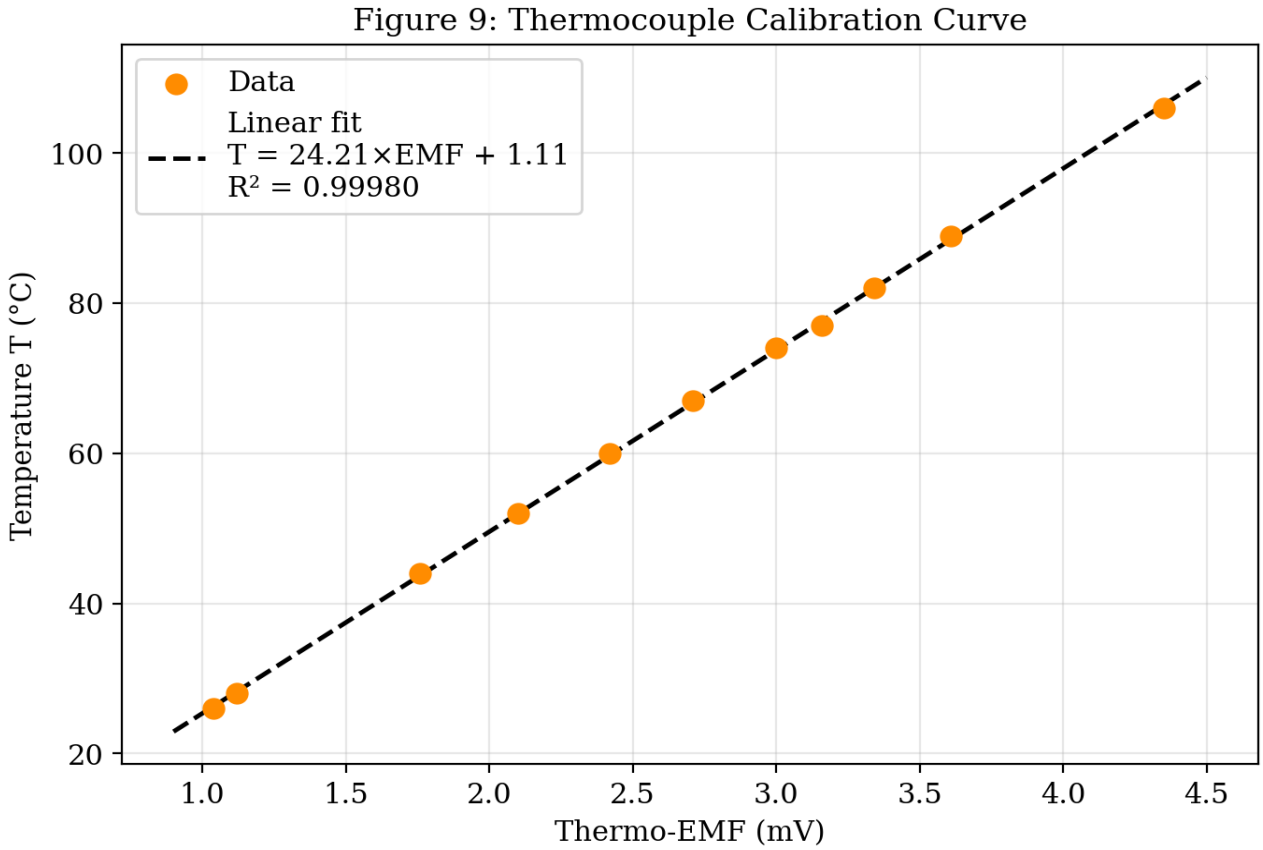

Figure 9: Thermocouple Calibration Curve — Temperature (°C) vs. Thermo-EMF (mV). Linear fit: T = 24.21 × EMF + 1.11, R^2 = 0.9998.